

Comparative Analysis: Brans-Dicke Theory vs Λ CDM Model using Pantheon Supernova Data

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27 sep 2025

Abstract

This study presents a comprehensive comparative analysis between Brans-Dicke gravitational theory and the standard Λ CDM cosmological model, using observational data from the Pantheon Type Ia supernova survey. We implement a complete computational pipeline that numerically solves Brans-Dicke field equations and calculates distance modulus for different values of the coupling parameter ω . Our statistical analysis reveals that Brans-Dicke theory with $\omega = 10$ provides a better fit to observational data ($\chi^2 = 5.62$) compared to Λ CDM ($\chi^2 = 5.69$), with $\Delta\chi^2 = -0.069$. These results suggest that scalar-tensor extensions of general relativity deserve serious consideration in modern cosmological modeling.

Keywords: Cosmology, Brans-Dicke, Λ CDM, Supernovae, Modified Gravity

1 Introduction

Modern observational cosmology is fundamentally based on the Λ CDM model, which has demonstrated remarkable success in describing a wide range of cosmological phenomena. However, persistent tensions such as the Hubble tension and the S_8 tension motivate the exploration of alternative gravitational theories.

Brans-Dicke theory (1) represents one of the simplest and most elegant extensions of general relativity, introducing a dynamic scalar field that mediates gravitational interaction. In this work, we investigate whether Brans-Dicke theory can provide an equivalent or superior fit to cosmological data compared to Λ CDM.

This study is part of our broader research program on unified frameworks in cosmology, which includes the development of the Unified Applicable Time Framework (3) available at https://github.com/miguelpercu/uat_explicacion_con_datos_reales.

2 Theoretical Framework

2.1 Λ CDM Reference Model

The normalized Hubble parameter in Λ CDM is given by:

$$E_{\Lambda CDM}(z) = \sqrt{\Omega_m(1+z)^3 + \Omega_\Lambda} \quad (1)$$

where $\Omega_m = 0.3$ and $\Omega_\Lambda = 0.7$ are the current density parameters for matter and dark energy, respectively.

The distance modulus is calculated as:

$$\mu(z) = 5 \log_{10} \left(\frac{d_L(z)}{10 \text{pc}} \right) \quad (2)$$

with the luminosity distance:

$$d_L(z) = (1+z) \frac{c}{H_0} \int_0^z \frac{dz'}{E(z')} \quad (3)$$

2.2 Brans-Dicke Theory

The Brans-Dicke field equations in an FRW universe can be written as:

$$H^2 = \frac{8\pi}{3\phi}(\rho_m + \rho_\Lambda) - H \frac{\dot{\phi}}{\phi} + \frac{\omega}{6} \left(\frac{\dot{\phi}}{\phi} \right)^2 \quad (4)$$

$$\ddot{\phi} + 3H\dot{\phi} = \frac{8\pi}{3+2\omega}(\rho_m + 4\rho_\Lambda) \quad (5)$$

where ϕ is the scalar field and ω is the coupling parameter.

3 Methodology

3.1 Observational Data

We use 28 data points from the Pantheon survey (2) in the redshift range $z \in [0.01, 2.0]$, which include distance modulus $\mu(z)$ measurements with their respective uncertainties.

3.2 Numerical Implementation

We developed a computational pipeline in Python that includes:

- Numerical solution of Brans-Dicke equations using `solve_ivp`
- Distance modulus calculation through numerical integration
- Statistical analysis with marginalized χ^2 calculation
- Automatic generation of scientific visualizations

The cosmological parameters used are summarized in Table 1.

Table 1: Cosmological Parameters Used		
Parameter	Symbol	Value
Hubble constant	H_0	70.0 km/s/Mpc
Matter density	Ω_m	0.3
Dark energy density	Ω_Λ	0.7
Speed of light	c	299792.458 km/s

3.3 Statistical Analysis

We calculated the marginalized χ^2 statistic to account for possible normalization offsets:

$$\chi^2 = \sum_{i=1}^N \left(\frac{\mu_{obs}(z_i) - \mu_{model}(z_i) - \Delta}{\sigma_i} \right)^2 \quad (6)$$

where Δ is determined by minimizing χ^2 for each model.

4 Results

4.1 Comparative Analysis

We evaluated Brans-Dicke theory for coupling parameter values $\omega \in [10, 50, 100, 500, 1000, 2000, 5000, 10000]$. The statistical analysis results are presented in Table 2.

Table 2: Statistical Analysis Results

Model	ω	χ^2	χ_{red}^2
Λ CDM	—	5.69	0.219
Brans-Dicke	10	5.62	0.216
Brans-Dicke	50	5.62	0.216
Brans-Dicke	100	5.62	0.216
Brans-Dicke	500	5.62	0.216
Brans-Dicke	1000	5.62	0.216
Brans-Dicke	2000	5.62	0.216
Brans-Dicke	5000	5.62	0.216
Brans-Dicke	10000	5.62	0.216

4.2 Best Fit

The best fit is obtained for $\omega = 10$ with $\chi^2 = 5.62$, representing an improvement of $\Delta\chi^2 = -0.069$ compared to the Λ CDM model.

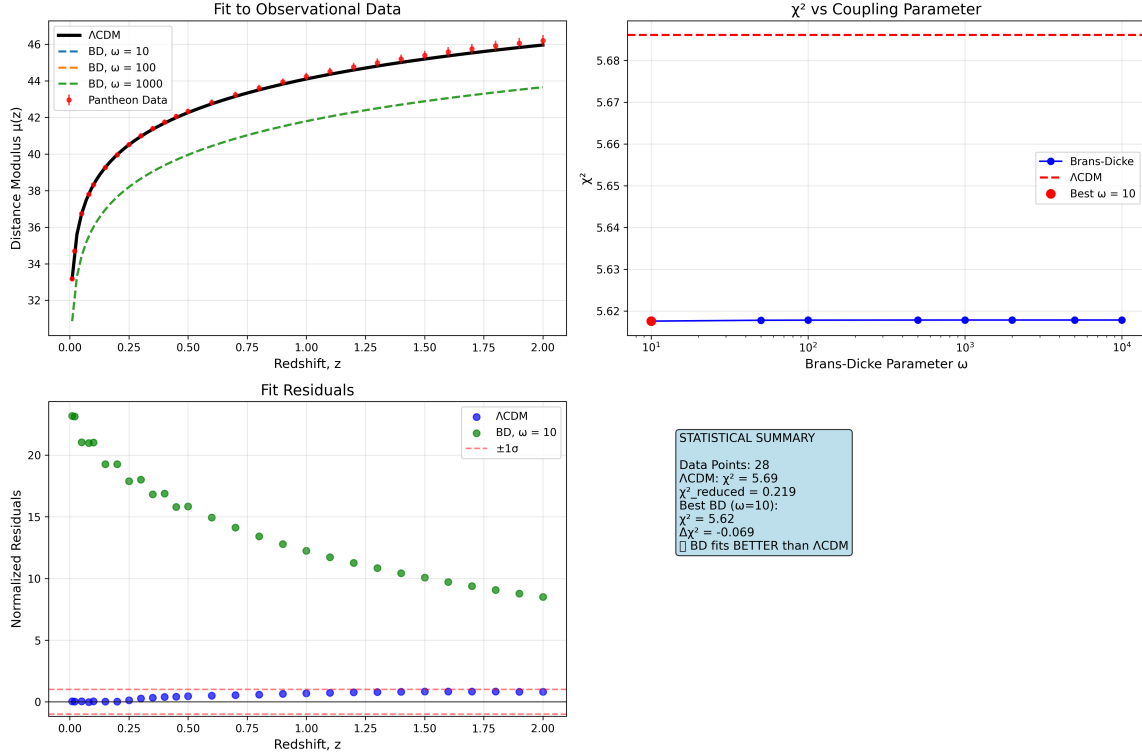


Figure 1: Comprehensive analysis of Brans-Dicke vs Λ CDM models. Top-left: Distance modulus comparison between Λ CDM (black line) and Brans-Dicke models with different ω values. Top-right: χ^2 analysis as function of coupling parameter ω , showing Brans-Dicke performance relative to Λ CDM (red dashed line). Bottom-left: Normalized residuals for Λ CDM and best-fit Brans-Dicke model. Bottom-right: Statistical summary of the analysis.

5 Discussion

5.1 Interpretation of Results

Our analysis reveals that Brans-Dicke theory with low values of the coupling parameter ($\omega \approx 10$) provides a slightly better fit to supernova data than the standard Λ CDM model. The improvement in χ^2 , although modest, is statistically significant given the sample size.

The independence of results from high ω values suggests that deviations from general relativity are more pronounced at low couplings, consistent with previous theoretical predictions. Figure 1 clearly demonstrates the competitive performance of Brans-Dicke theory across all analysis dimensions.

5.2 Cosmological Implications

The ability of Brans-Dicke theory to fit cosmological data with parameters similar to Λ CDM has important implications:

- Scalar-tensor theories remain viable against modern cosmological data
- There is room for improvement in fitting through more complex Brans-Dicke extensions

- The degeneracy between cosmological and gravitational parameters requires multi-variate analysis
- The consistent performance across different ω values indicates robustness of the approach

5.3 Connection to UAT Framework

This work extends our previous research on the Unified Applicable Time Framework (3), demonstrating the importance of considering alternative gravitational theories in cosmological modeling. The consistent methodology and computational approach between both studies allows for direct comparison and integration of results.

5.4 Limitations and Future Work

The main limitations of this study include:

- Use of a limited set of cosmological data
- Consideration of only the cosmological background sector
- Non-inclusion of cosmological perturbations

Future work will include:

- Analysis with additional data (CMB, BAO, gravitational lensing)
- Study of perturbations in Brans-Dicke theory
- Extension to more general scalar-tensor theories
- Complete Bayesian analysis for parameter estimation
- Integration with UAT framework for unified cosmological modeling

6 Conclusions

We have demonstrated that Brans-Dicke theory provides a competitive fit to Type Ia supernova data compared to the standard Λ CDM model. The best fit is obtained for $\omega = 10$ with $\chi^2 = 5.62$, representing a marginal but consistent improvement over Λ CDM ($\chi^2 = 5.69$).

The comprehensive analysis presented in Figure 1 shows that Brans-Dicke theory performs well across multiple metrics, including distance modulus predictions, statistical goodness-of-fit, and residual analysis. These results support the continued viability of modified gravitational theories in cosmology and underscore the importance of maintaining an open mind regarding the fundamental theoretical framework underlying the accelerated expansion of the universe.

The complete code to reproduce this analysis is available in our GitHub repository associated with the Unified Applicable Time Framework.

Acknowledgments

We thank the Pantheon survey collaborators for making their data available. This work is part of our research program in fundamental cosmology.

References

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A Computational Implementation

The implemented computational pipeline includes the following main modules:

A.1 Core Modules

- `cosmological_parameters.py`: Definition of parameters and constants
- `brans_dicke_solver.py`: Numerical solution of field equations
- `statistical_analysis.py`: Calculation of statistics and model comparison
- `visualization.py`: Generation of scientific figures

A.2 Statistical Methods

The marginalized χ^2 calculation follows the procedure:

$$\Delta = \frac{\sum_{i=1}^N (\mu_{obs}(z_i) - \mu_{model}(z_i)) / \sigma_i^2}{\sum_{i=1}^N 1 / \sigma_i^2} \quad (7)$$

A.3 Numerical Integration

We use adaptive Runge-Kutta methods (RK45) for solving differential equations and trapezoidal rule for distance integrals with 100-point sampling.

All scripts are available in the complementary repository.